Data Analytics Capstone

Gallstone Disease Risk Prediction Using Machine Learning

Interim Report

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QM640: Data Analytics Capstone

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GitHub repository

All raw data, processed data, code notebooks, and figures supporting this report are available in the public GitHub repository at:

https://github.com/prasun031087-afk/Capstone-Gallstone-Prediction

Introduction

Gallstone disease, also known as cholelithiasis, is a common digestive disorder caused by imbalances in bile components such as cholesterol or bilirubin, leading to hardened deposits in the gallbladder that may remain symptomless but can later cause severe pain, nausea, and jaundice. Affecting over 700 million people worldwide and increasing in prevalence, especially in urban areas due to lifestyle factors like poor diet and obesity, it is a major contributor to healthcare costs and emergency surgeries, highlighting the importance of early detection. The disease is influenced by several risk factors including age, gender, obesity, rapid weight loss, diabetes, genetics, and body composition measures such as visceral fat and muscle mass. Modern techniques such as bioelectrical impedance analysis, combined with biochemical blood markers, provide comprehensive data that can be used in machine learning models for improved risk prediction.

The dataset used in this study consists of clinical and bioimpedance data from 319 individuals, including 161 diagnosed cases of gallstones, with 38 features and a binary target variable. This structured dataset supports exploratory data analysis, statistical testing, and predictive modeling. Machine learning methods such as logistic regression and random forests are particularly suitable, offering both interpretability and accuracy. The study applies these techniques to generate reproducible insights and develop practical tools for early screening and risk assessment of gallstone disease.

Problem statement

Despite gallstone disease affecting hundreds of millions of patients worldwide, current clinical workflows continue to rely heavily on imaging studies that are typically ordered only after symptoms appear, which means at-risk individuals are often identified late in the disease course. While many traditional risk factors are well documented, there is no widely deployed, low-cost screening tool that combines routinely collected demographic, biochemical, and bioimpedance measurements to flag individuals who would benefit from earlier intervention. This project addresses that gap by examining whether a structured set of routinely available features can be used to estimate gallstone risk at the patient level using transparent and interpretable machine learning methods.

Purpose of the study

The purpose of this study is to use the publicly available UCI Gallstone-1 dataset to build and evaluate proof-of-concept supervised classification models that predict gallstone status from demographic, comorbidity, body composition, and serum biochemistry features. In parallel, the study aims to identify the strongest individual risk drivers through formal hypothesis testing, characterise how body composition varies across body mass index categories, and quantify whether obesity alone explains gallstone presence. The deliverable is a reproducible analytical pipeline that may inform the future design of clinical decision support systems and screening protocols rather than a tool intended for direct patient use.

Interim project status

Completed: data acquisition from the UCI Machine Learning Repository, defensive data cleaning, full exploratory data analysis with figures and tables, hypothesis testing for RQ1 through RQ3, baseline logistic regression and random forest model training, five-fold cross validation, held-out test evaluation, and feature importance analysis.

Scope and objectives

The scope of this study is limited to the Gallstone dataset (UCI Machine Learning Repository, Dataset ID 1150), which contains 319 patient records and thirty-eight clinical, biochemical, and bioimpedance features along with a binary target indicating gallstone presence. The analysis focuses on supervised binary classification and descriptive statistical exploration. The study does not seek to develop a clinical tool for direct patient use. Rather it produces validated proof of concept models and evidence-based observations that may inform the future design of clinical decision support systems and low-cost screening protocols. The four research questions along with the hypothesis that guide the study are stated below.

Research Question 1 (RQ1): What patterns exist in the distribution of clinical, biochemical, and bioimpedance features of the Gallstone dataset, and how do key variables differ between patients with gallstones and patients without gallstones?

Null Hypothesis (H_o)

For every continuous clinical, biochemical, and bioimpedance feature (including body mass index, total body fat ratio, visceral fat rating, glucose, low density lipoprotein, high density lipoprotein, alanine aminotransferase, and C reactive protein), the distribution is the same in patients with gallstones and patients without gallstones. For every categorical feature (including gender, comorbidity count, coronary artery disease, hypothyroidism, hyperlipidemia, and diabetes mellitus), gallstone status is independent of that feature.

Alternative Hypothesis (H_1)

At least one continuous feature shows a statistically significant difference in its distribution between patients with gallstones and patients without gallstones (Mann–Whitney U test, $p < 0.05$), or at least one categorical feature shows a statistically significant association with gallstone status (chi-square test of independence, $p < 0.05$).

Research Question 2 (RQ2): How do body composition metrics such as total body fat ratio, visceral fat rating, lean mass percentage, and muscle mass vary across body mass index categories ranging from underweight to obese, and what observable trends emerge as body mass index increases?

Null Hypothesis (H_0)

No observable monotonic trend is expected between body mass index category and body composition metric values, and the median values of the body composition metrics are expected to look similar across the four body mass index categories of underweight, normal, overweight, and obese when viewed through descriptive plots and summary statistics.

Alternative Hypothesis (H_1)

At least one body composition metric shows an observable monotonic trend, either positive or negative Spearman rank correlation, with the body mass index category, or its median value differs visibly across the four categories as revealed by exploratory box plots, violin plots, and median heatmaps.

Research Question 3 (RQ3): Is there a statistically significant association between obesity status, defined as a body mass index of thirty kilograms per square meter or higher, and the presence of gallstones in this patient population?

Null Hypothesis (H_0):

The proportion of patients with gallstones in the obese group is equal to the proportion of patients with gallstones in the non-obese group ($p_1 = p_2$).

Alternative Hypothesis (H_1):

The proportion of patients with gallstones in the obese group is not equal to the proportion of patients with gallstones in the non-obese group ($p_1 \neq p_2$).

Research Question 4 (RQ4): Can simple machine learning classification models accurately predict whether a patient has gallstones based on their clinical, biochemical, and bioimpedance features, and which features are the most important drivers of this prediction?

Null Hypothesis (H_0):

The trained classification model performs no better than chance in predicting gallstone status, that is the population area under the receiver operating characteristic curve is less than or equal to 0.50, and equivalently the model mean cross validated area under the curve is not statistically distinguishable from 0.50.

Alternative Hypothesis (H_1):

The trained classification model performs better than chance, that is the population area under the receiver operating characteristic curve is greater than 0.50.

Literature Survey

Eleven sources, six clinical or methodological and five software or statistical, inform the design of this study. Each source contributes to the conceptual, methodological, or analytical framework of the capstone.

Literature review approach

Sources for this study were identified through searches of PubMed, Google Scholar, and the UCI Machine Learning Repository using keyword combinations such as "gallstone disease," "cholelithiasis," "risk factors," "bioimpedance," "machine learning classification," "logistic regression," "random forest," "SMOTE," and "false discovery rate." Inclusion criteria favoured peer-reviewed clinical reviews on gallstone epidemiology, foundational methodological papers on the algorithms and statistical procedures used in the analysis, and computational toolkits required to reproduce the work.

Summary of key literature

Gallstone disease research is grounded in established epidemiological and clinical frameworks. Lammert, F. et al. (2016) provide a widely cited consensus on gallstones. They describe epidemiology, pathophysiology, and the four F risk factors. This supports the importance of early and non-invasive risk stratification. Stinton, L. M. and Shaffer, E. A. (2012) highlight obesity, diabetes, rapid weight loss, and lipid abnormalities as major modifiable risks. These findings inform the feature selection and research questions in this study.

The dataset used originates from Esen, B. et al. (2024). They applied logistic regression, k nearest neighbors, and support vector machines to a cohort of 319 patients. Their reported AUC of about 0.84 provides a useful benchmark for evaluating model performance in this study.

Methodologically, Chawla, N. V. et al. (2002) introduce the SMOTE technique. This method is demonstrated to address potential class imbalance in future datasets. Breiman, L. (2001) establishes the foundation for Random Forest. This supports its selection because it can model non-linear relationships and provide interpretable feature importance. Cohen, J. (1988) informs the study sample size and effect size calculations using established conventions such as Cohen w equals 0.30 and f squared equals 0.15.

From a computational perspective, Pedregosa, F. et al. (2011) provide the basis for the scikit-learn implementation used for preprocessing, model building, and evaluation. McKinney, W. (2010) supports the use of pandas for data manipulation and exploratory analysis. Statistical testing relies on Virtanen, P. et al. (2020) through SciPy. This enables non-parametric and contingency based analyses.

Public health standards are guided by the World Health Organization (2021). This source defines BMI classifications used in the study. Finally, multiple testing correction is handled using the false discovery rate procedure introduced by Benjamini, Y. and Hochberg, Y. (1995). This ensures robust statistical inference.

Table 1

Literature relevance matrix

Author (Year)	**Domain / Context**	**Dataset / Setting**	**Method(s)**	**Key Findings**	**Supports RQ / Decisions**
Lammert et al. (2016)	Clinical review of gallstones	Global epidemiology	Narrative consensus review	Defines four-F risk factors and epidemiology of cholelithiasis	Frames RQ1 and motivates risk stratification (RQ4)
Stinton & Shaffer (2012)	Epidemiology of gallbladder disease	Population-level evidence	Systematic review	Obesity, diabetes, lipid abnormalities are major modifiable risks	Justifies feature set for RQ2, RQ3
Esen et al. (2024)	Gallstone risk prediction	319 patients, Ankara cohort	LR, k-NN, SVM	Reported AUC ≈ 0.84 with bioimpedance + biochemistry features	Benchmark for RQ4; provides the dataset

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	Supports RQ / Decisions
Breiman (2001)	Statistical learning	Methodological	Random Forest theory	Ensemble of trees with feature importance	Model choice for RQ4
Chawla et al. (2002)	Imbalanced learning	Methodological	SMOTE oversamplin g	Synthetic minority oversampling improves rare-class recall	Mitigation strategy for class imbalance in RQ4
Cohen (1988)	Statistical power analysis	Methodological	Effect-size frameworks	Tabled values for w, f^2	Underpins sample-size calculations
Pedregosa et al. (2011)	Machine learning toolkit	Software	scikit-learn library	End-to-end ML pipelines in Python	Implementati on engine for RQ4
McKinney (2010)	Data analysis toolkit	Software	pandas library	Tabular data manipulation in Python	Used throughout RQ1, RQ2 EDA
Virtanen et al. (2020)	Scientific computing	Software	SciPy library	Statistical tests, distributions, optimisation	Hypothesis tests for RQ1, RQ3
Benjamini & Hochberg (1995)	Multiple testing	Methodological	False discovery rate	Controls expected proportion of false positives	Adjusts the 38 simultaneous tests in RQ1
WHO (2021)	Public health standards	Global guidelines	BMI classification	Underweight / normal / overweight / obese cutoffs	Defines BMI categories for RQ2, RQ3

Sample size calculation

To ensure that the study is powered adequately to detect meaningful effects and to validate model performance with statistical confidence, minimum sample sizes were calculated for each research question using established methods from statistical power analysis and confidence interval estimation. All calculations assume a two-tailed significance level of $\alpha = 0.05$ and, where applicable, a desired statistical power of $1 - \beta = 0.80$.

RQ1: Confidence Interval for a proportion: A confidence interval approach was used to estimate the proportion of patients with gallstones in the dataset within a margin of error of 0.055. Using a conservative estimated proportion of 0.50, which gives the largest required sample, the standard sample size formula for a proportion is shown below:

$$n = \frac{Z^2 . p . (1 - p)}{e^2} = \frac{(1.96)^2 \times 0.50 \times (1 - 0.50)}{(0.055)^2} = 317.32 \approx 318$$

RQ2: Confidence Interval for a mean: A confidence interval approach was used to estimate the mean body mass index value within each body composition category with a margin of error of 0.60 kilograms per square meter and an estimated standard deviation of 5.00 kilograms per square meter. The formula is given below:

$$n = \left(\frac{Z . \sigma}{e}\right)^2$$

where Z equals 1.96 for ninety five percent confidence, sigma equals 5.00 kilograms per square meter for an estimated standard deviation of body mass index drawn from clinical knowledge of this patient population, and e equals 0.60 kilograms per square meter for the desired margin of error.

After substituting the values into the formula:

$$n = \left(\frac{1.96 \times 5.0}{0.60}\right)^2 = 266.78 \approx 267$$

RQ3: Chi-square power analysis: This involves a formal hypothesis test comparing gallstone rates between two groups defined by obesity status. The goal is to enroll enough patients so the test can detect a real difference if one exists. The formula is given below.

$$n = \left(\frac{z_{\alpha/2} + z_\beta}{\omega}\right)^2$$

where $z_{\alpha/2}$ = 1.96 (critical value at α= 0.05, two tailed), z_β = 0.84 (critical value for 80% power), and ω = 0.30 (Cohen's w, a medium effect size for a chi-square test).

$$n = \left(\frac{1.96 + 0.84}{0.30}\right)^2 = 87.07$$

Hence, 87 patients per group is required, which gives 174 total samples across both obesity groups.

RQ4: Logistic regression power analysis: A logistic regression power analysis can be used. The minimum sample size needs to account for both the desired effect size and the number of predictors. The formula used comes from Cohen framework for multiple regression:

$$n = \left(\frac{L}{f^2}\right) + u + 1$$

where L = 7.85 is a tabled non-centrality parameter for power = 0.80 at alpha = 0.05, f^2 = 0.15 is a medium effect size for regression, and u = 38 is the number of predictor variables in the dataset.

After substituting the values into the formula:

$$n = \left(\frac{7.85}{0.15}\right) + 38 + 1 \approx 92$$

A minimum of 92 observations is required to build a reliable logistic regression model with 38 predictors, ensuring sufficient power to detect a medium effect size.

Table 2

Summary of Sample Size Calculations

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1	Confidence Interval (Estimating proportion of gallstone cases)	α = 0.05, Confidence level = 95 percent, Z = 1.96, percent so Z = 1.96, Estimated proportion p = 0.50, Margin of error e = 0.055	$\mathrm{N} = 318$
RQ2	Confidence Interval (Estimating mean body mass index)	α = 0.05, Confidence level = 95 percent, Z = 1.96, Standard deviation sigma = 5.00 kilograms per square meter, Margin of error e = 0.60 kilograms per square meter	$\mathrm{N} = 267$
RQ3	Power Analysis (Chi-square test for independence)	α = 0.05, z_α over 2 = 1.96, Power 80 percent i.e. z_β = 0.84, Effect size ω = 0.30	$\mathrm{N} = 174$
RQ4	Power Analysis (Logistic Regression, Cohen f^2)	L = 7.85 for Alpha = 0.05, Power = 0.80, f^2 = 0.15 medium effect size, u = 38 predictor variables	$\mathrm{N} = 92$

The final sample size for this study is chosen as the maximum of the minimum values computed across all four research questions, which is 318. The Gallstone dataset contains 319 total patient records with mostly complete feature data. Since 319 meets the required minimum of 318, the dataset is sufficient to support all planned analyses at the targeted precision and power levels.

Data Description

The dataset used in this study is the Gallstone dataset published on the University of California Irvine Machine Learning Repository. It is publicly accessible via the UCI repository at https://archive.ics.uci.edu/dataset/1150/gallstone-1. The dataset is provided as a single comma separated values file and can be loaded directly into Python using the pandas library (McKinney, 2010). It is also fully compatible with the scikit learn machine learning library (Pedregosa et al., 2011), which will be used to build and evaluate all classification models in this study. The dataset was collected at Ankara VM Medical Park Hospital and originally released by Esen et al. (2024). Each patient record carries a binary label that indicates whether gallstones were detected during the imaging examination.

Dataset Overview

Number of records (rows): 319 patient records

Number of variables (columns): 39 (38 predictors + 1 binary target)

Time period: Cross-sectional cohort collected at Ankara VM Medical Park Hospital and released through UCI in 2024

Unit of analysis: Individual patient

Target variable: Gallstone Status: binary (1 = confirmed gallstone case, 0 = control patient without gallstones)

Data dictionary

Table 3

Data dictionary in the Gallstone-1 dataset, grouped by role.

Variable Name	Definition	Datatype	Allowed Values/Range	Missing Values Handling	Notes
Gallstone Status	Target variable, Gallstones present (0), and absent (1)	Binary	0 = absent, 1 = present	None observed	Target variable
Age	Age of the person	Integer	Adult range (years, integer)	None observed	Demographics

Gender	Gender of the person	Categorical	0 = Male, 1 = Female	None observed	Demographics
Comorbidity	Concomitant diseases	Categorical	Integer count (0 to several)	None observed	Comorbidity
Coronary Artery Disease	Cardiovascular disease	Binary	0 / 1	None observed	Comorbidity
Hypothyroidism	Underactive thyroid gland	Binary	0 / 1	None observed	Comorbidity
Hyperlipidemia	High levels of fat in the blood	Binary	0 / 1	None observed	Comorbidity
Diabetes Mellitus (DM)	High blood sugar	Binary	0 / 1	None observed	Comorbidity
Height	Height is the length	Integer	Adult range (cm, integer)	None observed	Anthropometry
Weight	Body weight	Continuous	Adult range (kg)	None observed	Anthropometry
Body Mass Index (BMI)	Weight for height ratio	Continuous	Clinical range (kg/m^2)	None observed	Anthropometry
Total Body Water (TBW)	Total water in the body	Continuous	Bioimpedance range (L)	None observed	Bioimpedance
Extracellular Water (ECW)	It is extracellular water	Continuous	Bioimpedance range (L)	None observed	Bioimpedance
Intracellular Water (ICW)	It is intracellular water	Continuous	Bioimpedance range (L)	None observed	Bioimpedance
Extracellular Fluid/Total Body Water (ECF/TBW)	Extracellular water content	Continuous	0–1 (ratio)	None observed	Bioimpedance
Total Body Fat Ratio (TBFR)	Total fat content (units: %)	Continuous	0–100 %	None observed	Bioimpedance
Lean Mass (LM)	Lean body mass (units: %)	Continuous	0–100 %	None observed	Bioimpedance
Body Protein Content (Protein)	Essential building blocks for the body (units: %)	Continuous	0–100 %	None observed	Bioimpedance
Visceral Fat Rating (VFR)	Visceral organ fat level	Integer	Integer (typical 1–30)	None observed	Bioimpedance
Bone Mass-Bone Mass (BM)	Bone weight	Continuous	Bioimpedance range (kg)	None observed	Bioimpedance

Muscle Mass (MM)	Amount of muscle	Continuous	Bioimpedance range (kg)	None observed	Bioimpedance
Obesity	Excessive adiposity (units: %)	Continuous	0–100 %	None observed	Bioimpedance
Total Fat Content (TFC)	Total fat amount	Continuous	Bioimpedance range (kg)	None observed	Bioimpedance
Visceral Fat Area (VFA)	Inner adipose tissue area	Continuous	Bioimpedance range (cm^2)	None observed	Bioimpedance
Visceral Muscle Area (VMA)	Inner muscle area (units: kg)	Continuous	Bioimpedance range (kg)	None observed	Bioimpedance
Hepatic Fat Accumulation (HFA)	Accumulation of fat in the liver	Categorical	Ordinal grade (0–4)	None observed	Imaging
Glucose	Blood sugar	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Total Cholesterol (TC)	It is total cholesterol	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Low Density Lipoprotein (LDL)	Bad cholesterol	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
High Density Lipoprotein (HDL)	Good cholesterol	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Triglyceride	Type of fat found in the blood	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry
Aspartat Aminotransferaz (AST)	Type of liver enzyme	Continuous	Clinical reference range (U/L)	None observed	Serum biochemistry
Alanin Aminotransferaz (ALT)	An enzyme related to the liver	Continuous	Clinical reference range (U/L)	None observed	Serum biochemistry
Alkaline Phosphatase (ALP)	Type of liver and bone enzyme	Continuous	Clinical reference range (U/L)	None observed	Serum biochemistry
Creatinine	Kidney function indicator	Continuous	Clinical reference range (mg/dL)	None observed	Serum biochemistry

Glomerular Filtration Rate (GFR)	Kidney filtration rate	Continuous	Clinical reference range	None observed	Serum biochemistry
C-Reactive Protein (CRP)	Inflammation indicator	Continuous	≥ 0 (mg/L)	None observed	Serum biochemistry
Hemoglobin (HGB)	Protein in the blood that carries oxygen	Continuous	Clinical reference range (g/dL)	None observed	Serum biochemistry
Vitamin D	Essential vitamin for bone health	Continuous	Clinical reference range (ng/mL)	None observed	Serum biochemistry

The dataset includes 319 patient records with 39 variables in total, covering demographic, comorbidity, body composition, biochemical, and outcome information. The primary outcome variable is Gallstone Status, where 1 indicates a confirmed gallstone case and 0 indicates a control patient without gallstones. Out of 38 predictors, seven are treated as categorical or ordinal (Gender, Comorbidity, Coronary Artery Disease, Hypothyroidism, Hyperlipidemia, Diabetes Mellitus, and Hepatic Fat Accumulation) and thirty-one are continuous.

GitHub data availability

The raw and processed datasets are available in the GitHub repository under:

Raw data: data/raw/dataset-uci.xlsx

Processed / clean data: data/raw/dataset-uci.xlsx

Code / notebooks: notebooks/QM640_Data_Analytics_Capstone.ipynb

Analysis

Data cleaning

Three defensive data cleaning steps were applied before modeling. The dataset was checked for duplicates, incorrect labels, and unrealistic values, with no issues found. Although no missing data existed, an imputation pipeline was created for future use, applying median imputation for continuous variables and mode imputation for categorical ones. Continuous features were standardized for logistic regression, while the tree-based model used unscaled data. All preprocessing steps were implemented using scikit-learn Pipelines to prevent any information from the test set from influencing the training process during cross validation (Pedregosa et al., 2011).

Table 4

Data cleaning log.

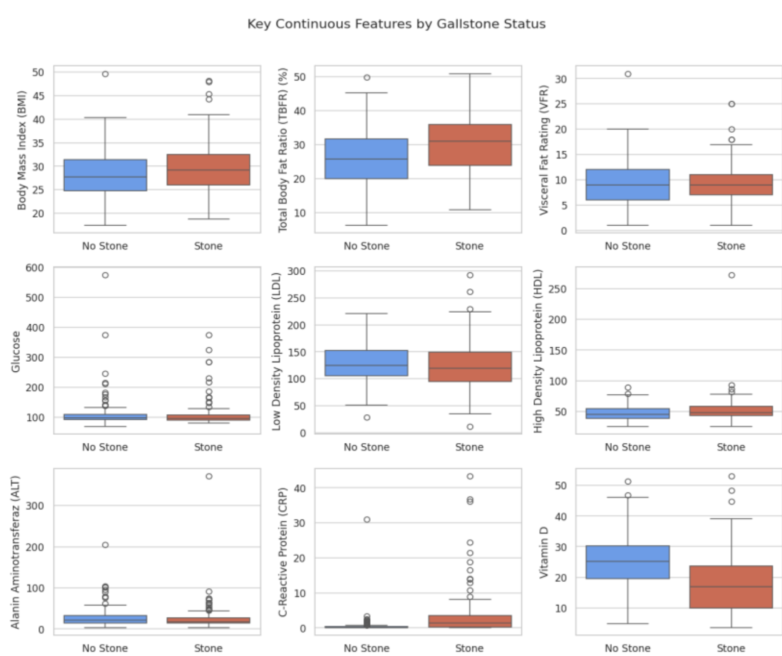
Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Missing values	All predictors (none observed in current data)	pandas isnull() audit, column-wise null counts	Median imputation for continuous, mode imputation for categorical, embedded in Pipeline	Robust to skewness and prevent leakage in future datasets
Outliers	Continuous bioimpedance and biochemistry features	Visual box-plot inspection	Retained (values may represent high-risk patients)	Clinical extremes are informative for risk modelling
Duplicates	Patient records	Row-level duplicate scan on full feature vector	None required (no duplicates found)	Preserves sample size of 319 patients
Implausible values	Anthropometry (Height, Weight, BMI)	Range checks against clinical plausibility	None required	All values within plausible adult ranges
Feature scaling	All 31 continuous predictors	Documented model-specific requirement	StandardScaler applied for logistic regression only	Tree-based models are scale-invariant

Exploratory Data Analysis

The exploratory data analysis was structured around the four research questions. The cohort was nearly perfectly balanced, with 158 gallstone cases and 161 controls, which minimized the need for resampling techniques. Figure 1 presents comparative box plots of nine routinely measured variables. CRP, total body fat ratio, and visceral fat rating shift upward in gallstone cases, while vitamin D and HDL-cholesterol shift downward. The gallstone group exhibited substantially higher levels of C reactive protein, total body fat ratio, and visceral fat rating, along with noticeably lower levels of vitamin D and high-density lipoprotein cholesterol, indicating a chronic inflammatory and fat dominant phenotype. In addition, lean mass percentage decreased while total body fat ratio increased among gallstone cases, even though body mass index showed considerable overlap between the two groups.

Figure 1

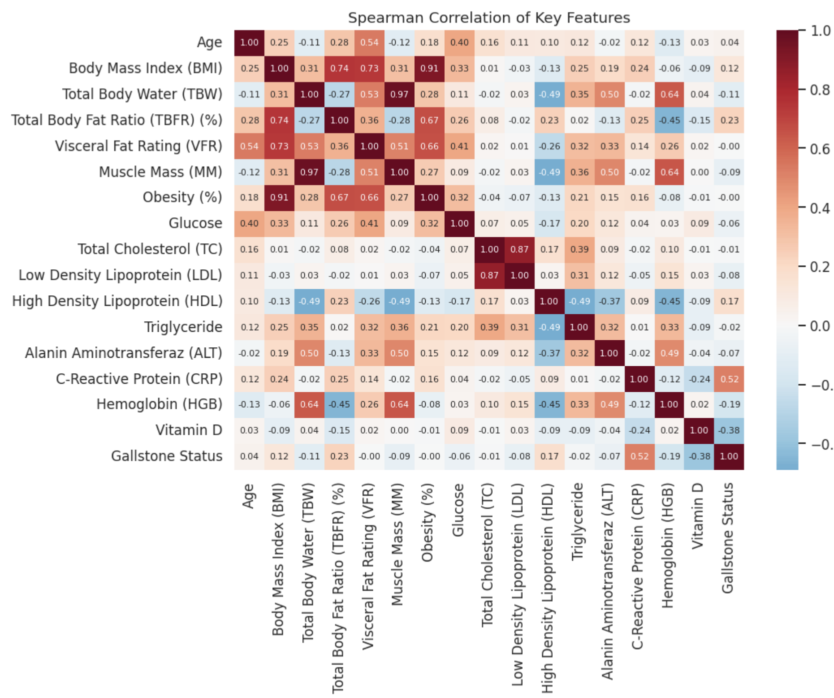
Distributions of nine key continuous features stratified by gallstone status.



Female patients, individuals with hyperlipidaemia, and those with hepatic fat accumulation exhibited the largest proportional differences in the stratified categorical comparisons, as shown in the accompanying notebook. Figure 2 illustrates the Spearman correlation heatmap for sixteen numeric variables alongside the outcome variable. Fat-related and inflammatory markers load positively; lean mass percentage and HDL are protective. Lean mass percentage demonstrated the strongest negative association with gallstone status, whereas C reactive protein and total body fat ratio showed the strongest positive associations.

Figure 2

Spearman correlation matrix of key numeric features and the outcome.



For the modeling component, the dataset is split into training and testing sets using an 80 to 20 ratio with stratified sampling to preserve class balance. Preprocessing includes feature standardization using zero mean and unit variance scaling. Model performance is assessed using five-fold cross validation on the training data to ensure reliable estimates, while the test set is reserved for a single final evaluation.

Hypothesis testing for research questions

RQ1 – Feature Distribution by Gallstone Status

Each of the 31 continuous predictors was evaluated using the Mann-Whitney U test to assess differences in distributions between gallstone and control groups. The seven categorical predictors were analyzed using the chi square test of independence, with Fisher exact test applied when any expected cell count was less than five. All p values were adjusted using the Benjamini Hochberg false discovery rate procedure at an alpha level of 0.05.

Among the continuous variables, 24 of 31 tests remained statistically significant after adjustment. Table 5 summarizes the ten most discriminative continuous variables, with C reactive protein, vitamin D, and aspartate aminotransferase showing the largest differences between groups, all with highly significant adjusted p values. Full test tables are reproduced in Appendix A.

Table 5

Top ten continuous features with the Benjamini–Hochberg adjusted p-values (Mann–Whitney U test).

Feature	Median (No stone)	Median (Stone)	Rank-biserial	Adj. p (BH)
C-Reactive Protein (CRP)	0.00	1.29	0.582	<0.001
Vitamin D	25.27	17.00	-0.438	<0.001
Aspartate Aminotransferase (AST)	20.00	16.50	-0.334	<0.001
Lean Mass (%)	73.93	68.79	-0.263	<0.001
Total Body Fat Ratio (%)	25.80	31.05	0.263	<0.001

Feature	Median (No stone)	Median (Stone)	Rank-biserial	Adj. p (BH)
Bone Mass	2.90	2.60	-0.252	<0.001
Hemoglobin	14.90	14.20	-0.221	0.002
Total Fat Content	20.10	24.75	0.223	0.002
Extracellular Water	17.50	16.55	-0.208	0.005
High Density Lipoprotein (HDL)	45.00	47.50	0.202	0.006

Table 6 summarizes the Chi-square and Fisher exact tests show that hepatic fat accumulation has the strongest association with gallstones, followed by hyperlipidemia and gender, all statistically significant after adjustment. Effect sizes are modest but meaningful. In contrast, diabetes mellitus, coronary artery disease, comorbidity count, and hypothyroidism are not significant, indicating weaker or no association in this dataset. Overall, liver fat and lipid-related factors emerge as the key categorical predictors of gallstone presence.

Table 6

Chi-square / Fisher exact results for the seven categorical predictors (RQ1). Significance assessed at adjusted $\alpha = 0.05$.

Feature	Test	χ^2 / statistic	dof	Cramer V	Adj. p (BH)	Significant
Hepatic Fat Accumulation	Chi-square	22.84	4	0.268	0.001	Yes
Hyperlipidemia	Fisher exact	6.42	1	0.142	0.012	Yes
Gender	Chi-square	6.91	1	0.147	0.020	Yes
Diabetes Mellitus	Chi-square	2.91	1	0.096	0.154	No
Coronary Artery Disease	Chi-square	2.07	1	0.081	0.211	No
Comorbidity count	Chi-square	2.99	3	0.097	0.459	No

Feature	Test	χ² / statistic	dof	Cramer V	Adj. p (BH)	Significant
Hypothyroidism	Fisher exact	0.42	1	0.036	0.502	No

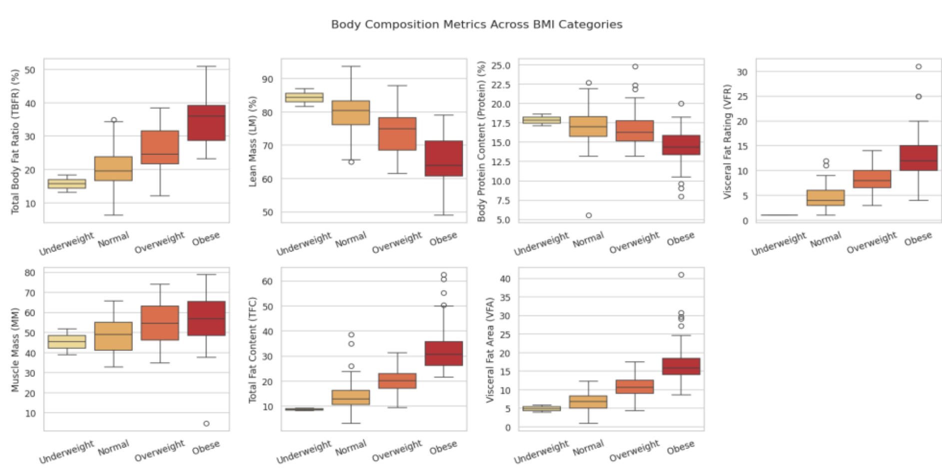
RQ2 – Body composition trends across BMI categories

Patients were classified into four World Health Organization categories based on body mass index: underweight defined as less than 18.5 kilograms per square meter, normal defined as 18.5 to 24.9, overweight defined as 25.0 to 29.9, and obese defined as 30 or greater. Most participants were concentrated in the normal and obese categories, while the underweight category included only one individual, which represents a limitation of the analysis.

Figure 3 shows box plot distributions of seven body composition measures across body mass index categories. A clear pattern is observed in which total fat content, visceral fat area, visceral fat rating, and total body fat ratio increase consistently with higher body mass index, whereas lean mass percentage and body protein percentage decrease.

Figure 3

Box plots of seven body-composition metrics across the four WHO BMI categories.



Spearman correlations shown in the Table 7 confirm these trends, with strong positive associations for visceral fat area and total fat content at 0.81, visceral fat rating at 0.72, and total body fat ratio at 0.66. Negative associations are observed for lean mass percentage at -0.66 and body protein percentage at -0.49.

Table 7

Spearman rank correlation between ordered BMI category and each body-composition metric.

Metric	Spearman ρ	p-value
Visceral Fat Area	0.808	<0.001
Total Fat Content	0.808	<0.001
Visceral Fat Rating	0.716	<0.001
Total Body Fat Ratio (%)	0.664	<0.001
Muscle Mass	0.298	<0.001
Body Protein (%)	-0.494	<0.001
Lean Mass (%)	-0.660	<0.001

RQ3 – Obesity versus gallstone presence

Patients were classified into two groups based on body mass index, with obesity defined as 30 or greater and non-obesity defined as less than 30, resulting in 121 obese and 198 non-obese individuals. The 2 × 2 contingency table comparing obesity status with gallstone presence is presented in Table 8, along with expected counts under the assumption of independence.

Table 8

Observed 2 × 2 contingency table for obesity status (BMI ≥ 30 vs BMI < 30) against gallstone status.

	No Stone	Stone	Total
Non-obese (BMI < 30)	104	94	198

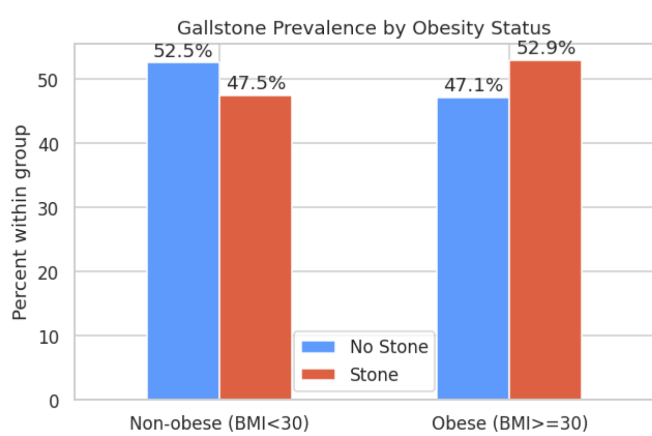
	No Stone	Stone	Total
Obese (BMI ≥ 30)	57	64	121
Total	161	158	319

A two tailed chi square test of independence indicated no statistically significant association between obesity and gallstone status, with chi square value of 0.68 and a p value of 0.410. The odds ratio was 1.24, and the 95 percent confidence interval ranged from 0.79 to 1.95, crossing one and therefore indicating no meaningful effect.

Figure 4 shows Gallstone prevalence within each obesity group. The two groups differ only modestly (47.5 % vs 52.9 %), consistent with the non-significant Chi-square result. Gallstone prevalence is slightly higher in obese individuals compared to non-obese individuals, but the difference is small. Non obese group shows a slightly higher proportion of no gallstone cases indicating minimal variation between groups.

Figure 4

Gallstone prevalence within each obesity group.



This indicates that a body mass index-based definition of obesity is a weak and non-significant predictor of gallstone status in this dataset.

In contrast, more detailed body composition measures examined in earlier analyses, particularly visceral fat area, lean mass percentage, and the inflammatory marker C reactive protein, demonstrate stronger and more meaningful associations.

EDA insight summary

Table 9

EDA insight summary.

Figure / Table Reference	What it Shows	Key Insight	Why it Matters (Link to RQ)	Decision / Next Step
Figure 1	Box plots of nine continuous features by gallstone status	CRP, total body fat ratio, and visceral fat rating shift upward in cases; vitamin D and HDL shift downward	Supports RQ1 - features differ between groups	Carry these features into modelling for RQ4
Figure 2	Spearman correlation heatmap of 16 features and outcome	Inflammatory and fat-related markers correlate positively with gallstone status; lean mass and HDL correlate negatively	RQ1, RQ4 - informs feature ranking	Compare with RF Gini importances
Figure 3	Body-composition metrics across WHO BMI categories	Fat-related metrics rise; lean and protein metrics fall as BMI increases	RQ2 - confirms monotonic trend	Use BMI category for descriptive segmentation only, not as direct predictor

Figure / Table Reference	What it Shows	Key Insight	Why it Matters (Link to RQ)	Decision / Next Step
Figure 4	Gallstone prevalence by obesity group	Modest 47.5 % vs 52.9 % difference, not significant	RQ3 - obesity alone is a weak predictor	Prefer fine-grained body-composition measures
Table 5	Top ten continuous features by FDR-adjusted p	CRP, vitamin D, AST, lean mass % most discriminative	RQ1, RQ4 - short-list of clinical drivers	Highlight these in screening recommendations
Table 6	Categorical predictors after FDR adjustment	Hepatic fat, hyperlipidemia, and gender significant	RQ1 - categorical drivers identified	Include in model and recommendations
Table 7	Spearman ρ of body composition vs BMI category	Visceral fat area and total fat content most strongly aligned with BMI	RQ2 - quantifies monotonic trends	Confirms direction for narrative
Table 8	2 × 2 contingency table for obesity vs gallstone	Odds ratio 1.24, 95 % CI [0.79, 1.95]	RQ3 - obesity alone insufficient	Move emphasis to detailed body composition

Modelling

Choice of Models with Justification

Two complementary classification models are developed for RQ4. Logistic regression serves as a transparent baseline, where standardised coefficients can be directly interpreted as log-odds contributions per standard deviation, supporting the interpretability requirement of a clinical screening tool. Random forest builds on this baseline by capturing non-linear relationships among body composition, biochemical, and demographic variables while still providing an interpretable ranking of feature importance (Breiman, 2001). The combination of a parametric linear model and a non-parametric ensemble allow the study to compare a strong, simple classifier against a flexible model on the same feature set, with both supported by the established benchmark of Esen et al. (2024).

Model 1 - Logistic Regression**:** Selected for interpretability, strong literature precedent in gallstone risk modelling, and direct comparability with the Esen et al. (2024) benchmark; sensitive to scale, so used with StandardScaler.

Model 2 - Random Forest**:** Selected to capture non-linear interactions, robustness to outliers, and built-in feature-importance ranking that complements logistic-regression coefficients; trained on unscaled features.

Features Included and Feature Engineering

All 38 features were retained. Categorical predictors were treated using their integer-coded representation in line with the source dataset, and continuous features were standardized for the logistic regression pipeline.

Table 10

Feature set and usage

Feature group	Original / Engineered	Type	Reason for Inclusion	Used in Model(s)
Demographics (Age, Gender)	Original	Continuous / Categorical	Established gallstone risk factors	LR, RF
Comorbidities (CAD, Hypothyroidism,	Original	Categorical / Ordinal	Clinical context and possible confounders	LR, RF

Feature group	Original / Engineered	Type	Reason for Inclusion	Used in Model(s)
Hyperlipidemia, DM, Comorbidity count)				
Anthropometry (Height, Weight, BMI)	Original	Continuous	Foundational measures for body-size effects	LR, RF
Bioimpedance compartments (TBW, ECW, ICW, ECF/TBW)	Original	Continuous	Capture fluid distribution differences	LR, RF
Body composition (Total Body Fat Ratio %, Lean Mass %, Body Protein %, Visceral Fat Rating, Bone Mass, Muscle Mass, Obesity %, Total Fat Content, Visceral Fat Area, Visceral Muscle Area)	Original	Continuous	Strongest EDA signals (Figures 1–3)	LR, RF
Imaging (Hepatic Fat Accumulation)	Original	Ordinal	Top categorical predictor in RQ1	LR, RF
Serum biochemistry (Glucose, Total Cholesterol, LDL, HDL, Triglyceride, AST, ALT, Alkaline Phosphatase, Creatinine, GFR, CRP, Hemoglobin, Vitamin D)	Original	Continuous	Major drivers from RQ1 hypothesis tests	LR, RF
BMI category, Obesity flag	Engineered (RQ2/RQ3 only)	Ordinal / Binary	Used for descriptive analyses; excluded from modelling to prevent leakage	Not used in modelling

RQ4 - Gallstone Status Prediction Using Machine Learning

Two complementary classification models were developed. Logistic regression serves as a transparent baseline, where standardized coefficients can be directly interpreted as log odds contributions per standard deviation. Random forest builds on this baseline by capturing non-linear relationships among body composition, biochemical, and demographic variables, while still providing an interpretable ranking of feature importance (Breiman, 2001).

All 38 features were retained, except for the composite indicators created for earlier research questions, namely body mass index category and obesity flag. These were excluded from the modeling process to prevent target leakage. The dataset was divided into 80 percent training and 20 percent testing sets using stratified sampling to maintain the class distribution. Hyperparameters were kept at default settings in scikit-learn, except for setting class weight to balanced and using 400 trees for the random forest model. Model performance was evaluated using two approaches: five-fold stratified cross validation on the training data with area under the curve reported as mean plus or minus standard deviation, and a final evaluation on the held-out test set.

Evaluation Metrics

Table 11 summarizes the model performance is evaluated using the standard metrics for a balanced binary classification problem with a clinical interpretation. All formulae are based on counts of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), or, for AUC, on the ranking of predicted probabilities.

Table 11

Evaluation metrics and formulae used in this study.

Metric	Formula	Interpretation	Used For
Accuracy	$\frac{(\mathrm{TP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})}$	Overall fraction of correct predictions	Balanced classification check
Precision	$\frac{\mathrm{TP}}{(\mathrm{TP}+\mathrm{FP})}$	Fraction of predicted positives that are true cases	Cost of false alarms

Metric	Formula	Interpretation	Used For
Recall (Sensitivity)	$\frac{\mathrm{TP}}{(\mathrm{TP}+\mathrm{FN})}$	Fraction of true cases correctly identified	Cost of missed cases
F1-score	$\frac{2 \times \mathrm{Recall} \times \mathrm{Precision}}{\mathrm{Recall}+\mathrm{Precision}}$	Harmonic mean balancing precision and recall	Imbalanced summary
ROC-AUC	$\int_0^1 TPR(FPR)dFPR$	Probability that a random case is ranked above a random control	Threshold-independent ranking
CV AUC (mean ± sd)	Mean and standard deviation of AUC across 5-fold stratified CV	Generalisation estimate on training data	Stability check

Preliminary Results

Interpretation of Hypotheses

Significant differences exist across features and body composition trends, obesity alone is not predictive, and both models demonstrate strong predictive performance.

RQ1 – Feature Distribution by Gallstone Status

Null hypothesis is rejected. Twenty four of the 31 continuous variables and three of the seven categorical variables show statistically significant differences between gallstone and control groups after false discovery rate adjustment, as reported in Tables 5 and 6. The global null hypothesis is therefore rejected, indicating that at least one feature differs between the two groups.

RQ2 – Body composition trends across BMI categories

Null hypothesis is rejected. All seven body composition measures demonstrate non-zero Spearman correlations with ordered body mass index category at p less than 0.001, as shown in Table 7.

Fat related measures increase consistently, while lean and protein measures decrease, supporting the alternative hypothesis.

RQ3 – Obesity versus gallstone presence

Null hypothesis is not rejected. The chi square test results show a value of 0.68 with a p value of 0.410, and the 95 percent confidence interval for the odds ratio ranges from 0.79 to 1.95, crossing 1, as presented in Table 8 and Figure 4. The null hypothesis that gallstone prevalence is equal between obese and non-obese individuals cannot be rejected. This suggests that detailed body composition measures provide stronger clinical insight than a binary obesity classification.

RQ4 - Gallstone Status Prediction Using Machine Learning

Null hypothesis is rejected. The cross validated area under the curve is 0.825 for logistic regression and 0.839 for random forest, both substantially above chance level. Test set area under the curve also exceeds 0.85 for both models. Therefore, the null hypothesis that classifier performance is equivalent to chance is rejected.

Preliminary Model Performance

Table 12 summarizes the performance metrics for both classification models, showing that each model performs well above chance level. The random forest model achieves higher values across accuracy, precision, recall, F1 score, and cross validated area under the curve, whereas logistic regression performs slightly better on test set area under the curve. This difference likely reflects variability due to the relatively small test sample size of 64 observations.

Table 12

Performance of Logistic Regression and Random Forest on the held-out test set (n = 64) and as 5-fold cross-validated estimates on the training set (n = 255).

Model	Accuracy	Precision	Recall	F1	Test AUC	CV AUC (mean ± sd)	CV Accuracy (mean ± sd)
Logistic Regression	0.766	0.793	0.719	0.754	0.881	0.825 ± 0.055	0.733 ± 0.051
Random Forest	0.797	0.806	0.781	0.794	0.867	0.839 ± 0.048	0.757 ± 0.063

Figure 5 presents the confusion matrices on the 64-patient held-out test set, indicating that both models produce a similar balance of false positives and false negatives, with no clear bias toward either class. Both models produce a balanced mix of false positives and false negatives.

Figure 5

Confusion matrices for Logistic Regression and Random Forest.

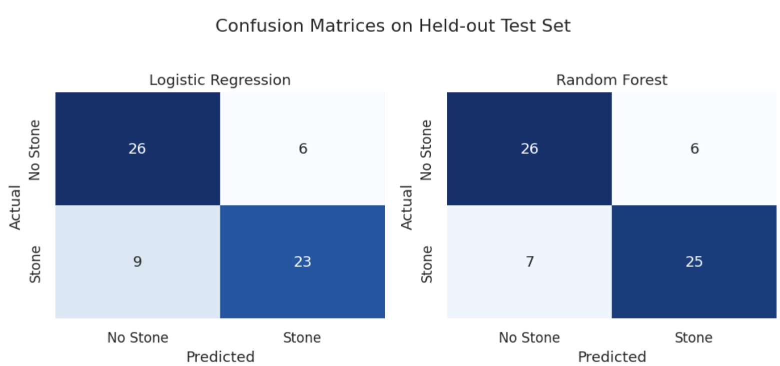
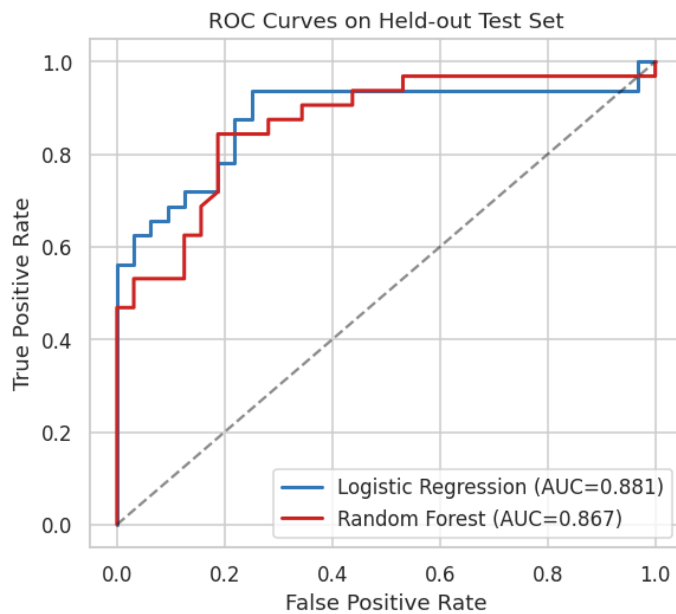


Figure 6 displays the receiver operating characteristic curves for both models. Both curves lie well above the diagonal, with logistic regression performing slightly better at lower false positive rates, while random forest shows stronger performance in the mid-range. The AUC is 0.881 for logistic regression and 0.867 for random forest, with both values clearly above the 0.50 chance level.

Figure 6

Receiver-operating characteristic (ROC) curves on the held-out test set.

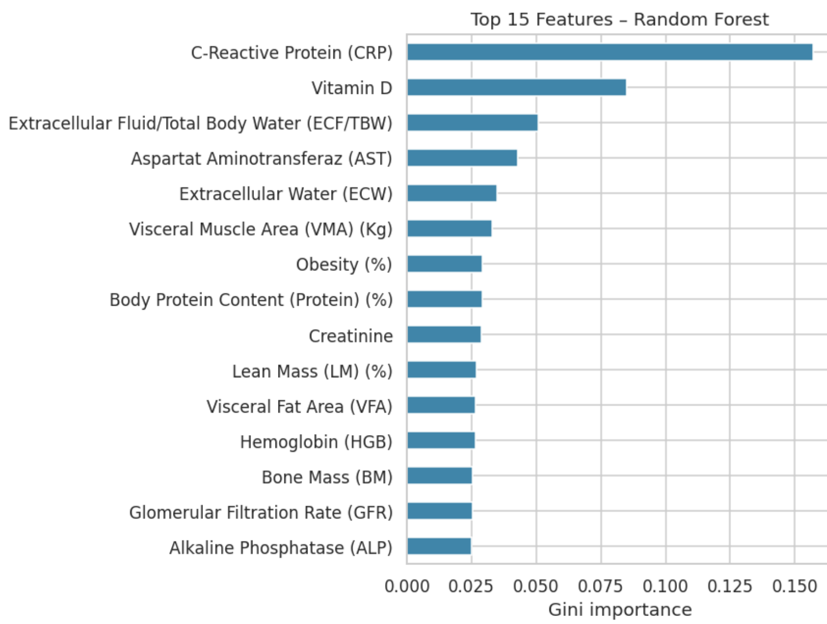


Feature importance and interpretation

Figure 7 shows the fifteen most important Random Forest features ranked by Gini importance. C-reactive protein dominates, followed by vitamin D, the extracellular-fluid ratio, AST, and a cluster of body-composition metrics. This ranking mirrors the hypothesis-testing output from RQ1 almost exactly, offering convergent evidence that systemic inflammation, vitamin D deficiency, and fat-versus-lean body composition are the most influential predictors of gallstone status in this cohort. The top five Logistic Regression standardised coefficients (hyperlipidemia, gender, CRP, bone mass, diabetes) point to the same short-list, giving clinicians a compact set of routine measurements to capture during outpatient screening.

Figure 7

Top 15 Random Forest feature importances (Gini). Inflammatory, vitamin D, and body-composition variables dominate the ranking.



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Appendix A – Reproducibility

All code, figures, and tabular results reported above were produced using the accompanying Jupyter or Colab notebook "Gallstone_Disease_Capstone.ipynb". In this Python notebook, the dataset file was uploaded to Google Drive to support the notebook execution. The notebook is self-contained: Section 2 fetches the dataset, Section 3 performs cleaning, Sections 4 to 7 implement RQ1 through RQ4, and Section 8 presents the final summary. The dataset file and the Python notebook are available in the GitHub repository. No local setup is required beyond the standard Python data science stack.